

MATH 114 - Computational Stochastics

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1 Overview

1.1 Introduction

Stochastic models describe real life phenomena like motion of the cell, shape of polymers stock prices, and more. What can we compute analytically and numerically from these models?

Key questions:

1. What is the distribution of the system?
2. How can we sample from the distribution?
3. How can we compute mean, variance, and higher moments of these systems?

Main topics:

1. Sampling RVs

2. Markov chain Monte Carlo (MCMC)
3. Stochastic differential equations (numerical & computation)

1.2 Sampling

- Generating "hard" RVs from "easy" RVs.
- Compute "hard" integrals.

ex Area of the unit circle, $A = \pi$

Let $U_1 \sim U(-1, 1)$ and $U_2 \sim U(-1, 1)$

Let $x = \begin{cases} 0 & \text{if } (u_1, u_2) \text{ outside circle} \\ 1 & \text{if } (u_1, u_2) \text{ inside circle} \end{cases}$

$\implies P(X = 1) = \frac{A}{4}, P(X = 0) = 1 - \frac{A}{4}$

$\mathbb{E}(X) = \frac{A}{4}$

After N trials we generate $X_1, X_2, \dots, X_n$ iid

By the strong law of large numbers

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N X_i = \frac{A}{4} = \frac{\pi}{4}$$

1.3 MCMC

Construct Markov chains to generate target distribution.

ex Ising model